# Digital Forensics (IY5609 / IY4609 / IY3609) Summative Assessments – CW1 Discussion of Moodle Quiz

Dr Christian Weinert

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# Reminder: Record Your Attendance!

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# Overview

- Q1: Legal and Procedural Aspects I
- Q2: Legal and Procedural Aspects II
- Q3: Digital Evidence I
- Q4: Digital Evidence II

# Q1: Legal and Procedural Aspects I

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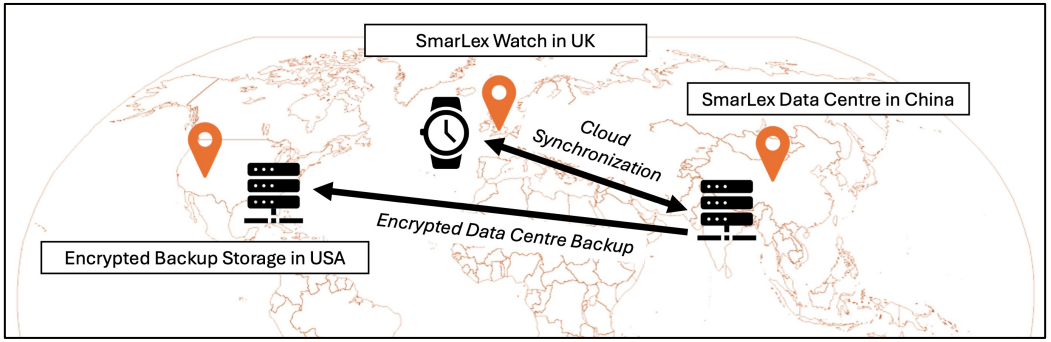
# Question (1/2)

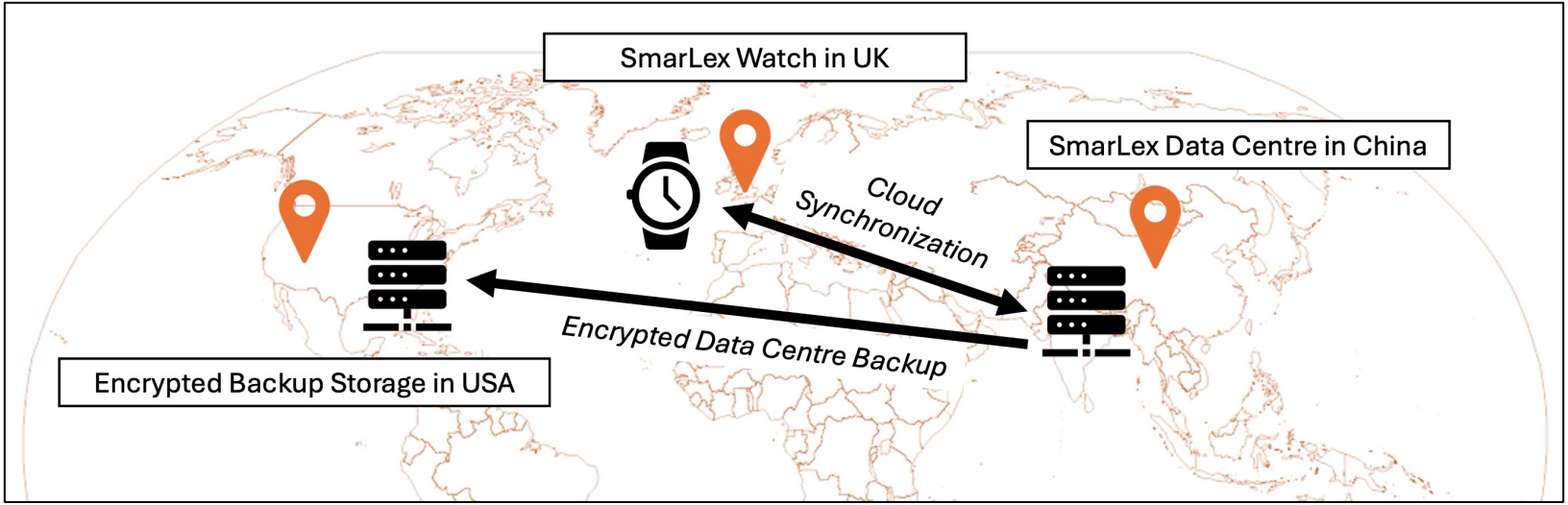
Caleb is investigating a kidnapping and homicide case where the victim's body was found in a nearby river. At the crime scene, he notices the victim is wearing a 5G-enabled "SmarLex" smartwatch, which usually records GPS history and thus could potentially reveal the kidnapper's location. Unfortunately, due to extended water exposure, the watch is no longer functional. However, it presumably has synced previous location data to the cloud.

A schematic overview of the synchronization process and the SmarLex cloud infrastructure is shown in the figure below.



# Question (2/2)

For each cloud storage location, identify and describe an appropriate legal procedure that Caleb can follow to request access to the victim's SmarLex account data. Furthermore, discuss whether Caleb is likely to obtain GPS data about the victim's last known locations in a timely manner.



# Model Answer

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SmarLex Data Centre in China:

- For the lack of a data access/sharing agreement with China, Caleb may need to initiate a Mutual Legal Assistance Treaty (MLAT) request [5].
- MLAT requests involve diplomatic channels where UK law enforcement submits a formal request to Chinese authorities, who then decide whether to comply based on their legal framework [5].

Encrypted Backup Storage in USA:

- Caleb can use the bilateral UK-US Data Access Agreement (DAA) to obtain data from the encrypted backup stored in the USA [5].
- Under the DAA, UK law enforcement can apply for an Overseas Production Order (OPO), which allows direct requests to US-based service providers [5].

It is unlikely that Caleb will obtain GPS data in time, as the MLAT process is slow, and while the DAA may allow quick access to the backup, the encryption renders it unusable [5].

# Q2: Legal and Procedural Aspects II

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# Question (1/2)

Jess is leading the investigation against the administrator of a dark web forum that was used to distribute illicit content. During a raid at the suspect's residence in Surrey, her team discovers the following server rack in the basement.

# Question (2/2)

Suggest how Jess and her team should proceed when collecting (digital) forensic evidence towards analyzing content stored on the servers. For this, outline five important steps, briefly explaining your rationale and how the continuity of evidence is preserved.

# Model Answer

1. Freeze and thoroughly document the scene (e.g., take photos/videos + make notes of serial numbers etc.) to preserve as much contextual information as possible and to prevent claims of evidence tampering [5].

2. Decide whether to attempt live acquisition (i.e., obtaining memory snapshots of the servers) to not lose access to key material and other valuable volatile data (e.g., active network connections); note that in case of counter-forensic techniques this might harm evidence [5].

3. Power off the servers and remove covers, documenting the internal configuration to again preserve contextual information and linking the hardware to the scene [5].

4. Remove storage media, documenting manufacturer details and configuration (especially order of drives in case of RAID), and place in anti-static bag to prevent damaging the electronics.

5. Place anti-static bag in envelope, seal + sign it together with a witness to ensure a clear chain of custody [5].

# Q3: Digital Evidence I

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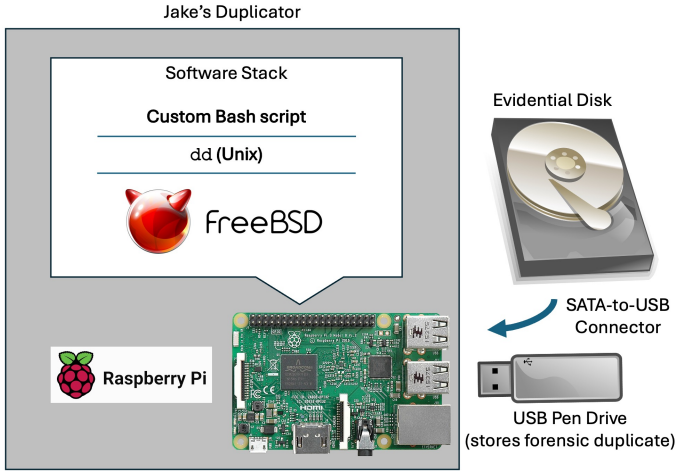
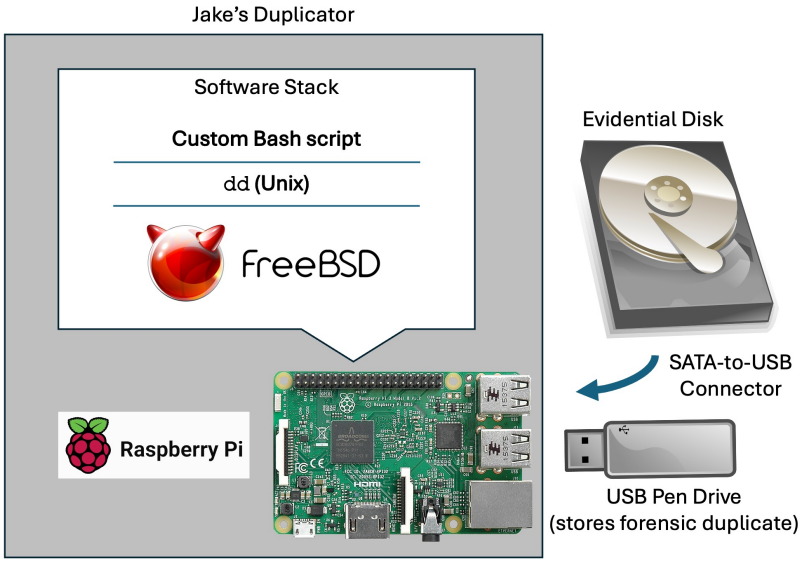
# Question (1/2)

Jake is a passionate engineer who runs a one-person manufacturing business specializing in custom electronic devices, ranging from embedded control systems to network security appliances. From a friend who works in a forensic lab, Jake learns that the lab is planning to invest significantly in a new generation of forensic duplication devices. Seeing a business opportunity, Jake quickly designs his own duplication device, hoping to secure the lab's contract. His planned hardware and software architecture is shown in the following diagram:



# Question (2/2)

Is Jake's plan a good idea? In particular, would analysis results based on forensic duplicates created with Jake's device be admissible as evidence in court? Discuss potential concerns regarding data integrity and how reliability could be demonstrated in court.



# Model Answer

The first major issue is that the SATA-to-USB connector does not physically block write commands to the evidential disk [5]. This could result in altering the original data, violating the first ACPO principle, as it cannot be guaranteed that the duplicator does not issue unintended write commands [5].

The second major issue is the use of an unvalidated system, which lacks independent verification of both hardware and software [5]. Proving reliability in court is therefore difficult, especially given the challenges of validating a general-purpose processor running a full BSD operating system with custom scripts [5].

As a result, forensic duplicates created with Jake's device would likely be inadmissible as evidence in court [5].

# Q4: Digital Evidence II

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# Question (1/2)

Keila is preparing a drive for forensic analysis, which was recovered from the laptop of a suspected hacker involved in a financial fraud case. In particular, the suspect is accused of emptying multiple bank accounts on the 24th of July 2024. Before starting forensic duplication, Keila reads out the drive's SMART status. She sees the following data:

Device Info | SMART Info | SMART History | Disk Self Test | Hidden Areas - HPA/DCO

Temperature — Records Found: 20 — Records Displayed: 20

| Date/Time | Value | Worst | Raw Value |
|---|---|---|---|
| Thu Jul 25 18:28:27 2024 | 37 | 45 | 73014444069 |
| Thu Jul 25 18:22:18 2024 | 37 | 45 | 73014444069 |
| Thu Jul 25 18:15:12 2024 | 37 | 45 | 73014444069 |
| Thu Jul 25 18:12:37 2024 | 37 | 45 | 73014444069 |
| Thu Jul 25 18:08:44 2024 | 37 | 45 | 73014444069 |
| Tue Jul 23 16:52:43 2024 | 37 | 45 | 73014444069 |
| Tue Jul 23 17:55:23 2024 | 37 | 45 | 73014444069 |
| Tue Jul 23 17:52:12 2024 | 37 | 45 | 73014444069 |
| Tue Jul 23 17:48:55 2024 | 36 | 45 | 73014444068 |
| Tue Jul 23 17:42:37 2024 | 37 | 45 | 73014444069 |
| Tue Jul 23 17:36:45 2024 | 37 | 45 | 73014444069 |
| Tue Jul 23 17:31:33 2024 | 37 | 45 | 73014444069 |
| Tue Jul 23 17:22:21 2024 | 36 | 45 | 73014444068 |
| Tue Jul 23 17:15:29 2024 | 37 | 45 | 73014444069 |
| Tue Jul 23 17:12:10 2024 | 37 | 45 | 73014444069 |
| Tue Jul 23 17:08:47 2024 | 37 | 45 | 73014444069 |
| Tue Jul 23 17:07:17 2024 | 37 | 45 | 73014444069 |

# Question (2/2)

Make two observations that might be relevant for the investigation. For each, explain its forensic significance. Furthermore, discuss whether Keila can fully trust the accuracy of the SMART data.

Device Info | SMART Info | SMART History | Disk Self Test | Hidden Areas - HPA/DCO

Temperature — Records Found: 20 — Records Displayed: 20

| Date/Time | Value | Worst | Raw Value |
|---|---|---|---|
| Thu Jul 25 18:28:27 2024 | 37 | 45 | 73014444069 |
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# Model Answer

Observation 1: The SMART history indicates that the drive was not operational at all on the 24th of July 2024 [5]. This could suggest that the suspect did not use the laptop during the alleged crime, potentially casting doubt on his involvement [5].

Observation 2: There is a timeline inconsistency in the SMART records (between the 23rd and 25th of July there is an entry for 23rd of July that has an earlier timestamp than previous entries) [5]. This may indicate system clock manipulation or an attempt to obfuscate forensic traces, possibly hiding malicious activities [5].

Reliability of SMART Data: Due to the lack of tamper resistance mechanisms, SMART data cannot be fully trusted, as a sufficiently skilled adversary could modify records within the drive's service area to manipulate forensic timelines [5].

# Common Mistakes

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# Common Mistakes (1/2)

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In general:

- Poor structure for responses
- Not reading questions properly / not working with the given data

Q1:

- Basing arguments solely on CLOUD Act
- Assuming additional servers in the UK
- Focus on data recovery from the damaged smart watch
- Not describing how the proposed legal procedure works

Q2:

- Just saying ”this helps to preserve the continuity of evidence" => how/why?

# Common Mistakes (2/2)

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Q3:

- Use of Daubert criteria to test evidence admissibility
- Questioning Jake's qualifications in handling evidence
- Suggesting issues due to USB pen drive storage architecture differing from evidential drive
- Blaming low compute power of Raspberry Pi
- Referencing ISO/IEC 17025:2017

Q4:

- Confusing raw temperature value with raw read error rate
- Assuming raw value differs from presented human-readable temperature values
- Unsubstantiated arguments about trustworthiness of SMART data

# Questions?

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